Heinrich-Heine-Universität Düsseldorf
Computational Cell Biology
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Applications of Transformer Networks in Bio- and Cheminformatics

Worksheet 4

Submission Deadline:	19. Mai 2025, 11:59 pm
Discussion of solutions:	20. Mai 2025, 12:30 - 14:00

Submission Instructions:

- Upload a Jupyter notebook with your solutions to the exercises that you solved on your own PC. Upload the Python scripts and log files for any code executed on the HPC.
- If you are submitting multiple files, zip them together.
- Submit your solutions by uploading the Jupyter notebook to https://uni-duesseldorf.sciebo.de/s/hCt1rTP23EeWmUC. The uploaded file should have the following filename: "lastname_studentID_worksheet4".

Excercise 4.1 *Gradient Boosting Hyperparameter Optimization* (50 Points)

For this task, you will need to familiarize yourself with the Python package hyperopt. You will perform a hyperparameter optimization for a gradient boosting model for predicting the optimal pH of proteins using the dataset from Worksheet 1. The input features will be the frequencies of all 1-mers and 2-mers, i.e. you will have input vectors of dimension 420. Implement the gradient boosting models using the Python package xgboost.

(a) Perform a hyperparameter optimization for a gradient boosting model (xgboost) using the hyperopt package. Search for the best set of hyperparameters among the following parameters with the following upper and lower bounds:

- Number of trees ("n_estimators"): Lower bound: 10, Upper bound: 500,
- Maximum tree depth: ("max_depth"): Lower bound: 1, Upper bound: 10,
- Learning rate: ("learning_rate"): Lower bound: 0.01, Upper bound: 0.5
- Min. child weight ("min_child_weight"): Lower bound: 1, Upper bound: 10,
- Lambda ("reg_lambda"): Lower bound: 0, Upper bound: 1,
- Alpha ("reg_alpha"): Lower bound: 0, Upper bound: 1.

Search for the best set of hyperparameters on the validation set by iterating over at least 200 different combinations of hyperparameters. Search for the set that leads to the lowest coefficient of determination R^2. If this takes too long on your PC, run the code on the HPC.

Hint: Consider that hyperopt tries to minimize a function but you want to maximize R^2.

(b) Select the set of hyperparameters that led to the best R^2 on the validation set in (a). Train a gradient boosting model with this set of hyperparameters on the training set and evaluate it on the test set, i.e., calculate MSE and R^2 for the test set.

(c) Follow the instructions from (a) and perform a hyperparameter optimization of a gradient boosting model but with Pfam domains as input features. You can use the data and solutions from Exercise 1.3 to obtain the input matrices.

Excercise 4.2 *ESM-2 embeddings for pH prediction* (50 Points)

(a) Compute ESM-2 (model: esm2_t6_8M_UR50D) protein embeddings for all sequences in the pH dataset. This should give you a single protein representation of dimension 320 for each protein sequence. For an example on how to compute these embeddings, see the Jupyter notebook *data/worksheet4/Get_ESM2_embeddings.ipynb*. If you cannot calculate the embeddings on your PC but instead need to use the HPC, you can find instructions in the same Jupyter notebook for how to move the ESM-2 model to the HPC and load it from a local file.

(b) Perform a hyperparameter optimization of a gradient boosting model as described in Exercise 4.1 (a) but with the ESM-2-embeddings as input features.

(c) Train a gradient boosting model with this set of hyperparameters on the training set and validate it on the test set, i.e., calculate MSE and R^2 for the test set.

Excercise 4.3 *Use of LLMs* (0 Points)

State for which exercise you have used LLMs (large language models) such as ChatGPT or GitHub Copilot. State which tools you have used and for which steps. This answer does not influence how many points you receive for your submission.